

## Data Science Jobs Analysis Project using SQL and Power BI

### About the Dataset:

The Data Science jobs dataset is from Kaggle. It contains real job postings scraped from Glassdoor, representing data science and data-related job listings across the United States. This dataset will serve as my practice and a project for my portfolio.

Link: <https://www.kaggle.com/datasets/rashikrahmanpritom/data-science-job-posting-on-glassdoor>

The steps I have taken for this project are Data Cleaning using SQL, which includes fixing, standardizing, and enriching the dataset for it to be ready for Exploratory Data Analysis (EDA). Then I performed EDA to find patterns and insights hidden in the dataset using SQL also. Lastly, the last step I did is to create a dashboard out of my cleaned dataset to visualize and to prepare the data for reporting using Power BI.

| index | job_title         | salary_estimate                | job_description    | rating | company_name           | location      | headquarters           | size                | founded | type_of_ownership       | industry                       | sector            | revenue                  | competitors           |
|-------|-------------------|--------------------------------|--------------------|--------|------------------------|---------------|------------------------|---------------------|---------|-------------------------|--------------------------------|-------------------|--------------------------|-----------------------|
| 0     | Sr Data Scientist | \$137K-\$171K (Glassdoor est.) | Description Th...  | 3.1    | Healthfirst 3.1        | New York, NY  | New York, NY           | 1001 to 5000 emp... | 1993    | Nonprofit Organization  | Insurance Carriers             | Insurance         | Unknown / Non-Ap...      | EmblemHealth, Unit... |
| 1     | Data Scientist    | \$137K-\$171K (Glassdoor est.) | Secure our Na...   | 4.2    | ManTech 4.2            | Chantilly, VA | Herndon, VA            | 5001 to 10000 em... | 1968    | Company - Public        | Research & Development         | Business Services | \$1 to \$2 billion (USD) | NULL                  |
| 2     | Data Scientist    | \$137K-\$171K (Glassdoor est.) | Overview An...     | 3.8    | Analysis Group 3.8     | Boston, MA    | Boston, MA             | 1001 to 5000 emp... | 1981    | Private Practice / Firm | Consulting                     | Business Services | \$100 to \$500 millio... | NULL                  |
| 3     | Data Scientist    | \$137K-\$171K (Glassdoor est.) | JOB DESCRIPT...    | 3.5    | INFICON 3.5            | Newton, MA    | Bad Ragaz, Switzerland | 501 to 1000 empl... | 2000    | Company - Public        | Electrical & Electronic Man... | Manufacturing     | \$100 to \$500 millio... | MKS Instruments, ...  |
| 4     | Data Scientist    | \$137K-\$171K (Glassdoor est.) | Data Scientist ... | 2.9    | Affinity Solutions 2.9 | New York, NY  | New York, NY           | 51 to 200 employees | 1998    | Company - Private       | Advertising & Marketing        | Business Services | Unknown / Non-Ap...      | Commerce Signals, ... |

**Figure 1. Snapshot of the Raw Dataset**

Looking at the raw table, I can already see several issues across almost all columns. The job title column contains overly specific titles that refer to the same role, for example, Analytics Manager and Analytics Manager - Data Mart refer to the same position. The salary estimate column stores values in a formatted string like \$137K-\$171K (Glassdoor est.) which needs to be parsed into usable numbers. The company name column contains the Glassdoor rating appended right after the name, for example, Healthfirst 3.1, which needs to be separated. The location and headquarters columns store both city and state together in one field separated by a comma. Some location values are just the full state name or the country itself (United States) instead of a city-state pair. There is also a company name with an encoding issue, bioMerieux, which appears to be a character encoding error from the import process.

Additionally, the dataset contains no duplicate records, so I did not need to remove any. However, I decided to keep null and missing values rather than removing rows, because I believe removing records would reduce the accuracy of the dataset. Nulls will be filtered out during EDA instead.

| index | cleaned_job_title | senior | min_salary_estimate | max_salary_estimate | average_salary | cleaned_company_name | rating | location_city | location_state | headquarters_city | headquarters_state | same_state | size                    | founded | type_of_ownership       |
|-------|-------------------|--------|---------------------|---------------------|----------------|----------------------|--------|---------------|----------------|-------------------|--------------------|------------|-------------------------|---------|-------------------------|
| 0     | Data Scientist    | 1      | 137                 | 171                 | 154.00         | Healthfirst          | 3.1    | New York      | NY             | New York          | NY                 | 1          | 1001 to 5000 employees  | 1993    | Nonprofit Organization  |
| 1     | Data Scientist    | 0      | 137                 | 171                 | 154.00         | ManTech              | 4.2    | Chantilly     | VA             | Herndon           | VA                 | 1          | 5001 to 10000 employees | 1968    | Company - Public        |
| 2     | Data Scientist    | 0      | 137                 | 171                 | 154.00         | Analysis Group       | 3.8    | Boston        | MA             | Boston            | MA                 | 1          | 1001 to 5000 employees  | 1981    | Private Practice / Firm |
| 3     | Data Scientist    | 0      | 137                 | 171                 | 154.00         | INFICON              | 3.5    | Newton        | MA             | Bad Ragaz         | Switzerland        | 0          | 501 to 1000 employees   | 2000    | Company - Public        |
| 4     | Data Scientist    | 0      | 137                 | 171                 | 154.00         | Affinity Solutions   | 2.9    | New York      | NY             | New York          | NY                 | 1          | 51 to 200 employees     | 1998    | Company - Private       |

**Figure 2.1 Snapshot of the Cleaned Dataset**

| industry                              | sector            | revenue                      | python | r_skills | sql_skills | spark | hadoop | tableau | excel | aws | azure | tensorflow | keras | pytorch | sas |
|---------------------------------------|-------------------|------------------------------|--------|----------|------------|-------|--------|---------|-------|-----|-------|------------|-------|---------|-----|
| Insurance Carriers                    | Insurance         | Unknown / Non-Applicable     | 0      | 0        | 0          | 0     | 0      | 0       | 0     | 1   | 1     | 0          | 0     | 0       | 0   |
| Research & Development                | Business Services | \$1 to \$2 billion (USD)     | 0      | 0        | 1          | 0     | 1      | 0       | 0     | 0   | 0     | 0          | 0     | 0       | 0   |
| Consulting                            | Business Services | \$100 to \$500 million (USD) | 1      | 1        | 0          | 0     | 0      | 0       | 1     | 1   | 0     | 0          | 0     | 0       | 0   |
| Electrical & Electronic Manufacturing | Manufacturing     | \$100 to \$500 million (USD) | 1      | 0        | 1          | 0     | 0      | 0       | 1     | 1   | 0     | 1          | 0     | 1       | 0   |
| Advertising & Marketing               | Business Services | Unknown / Non-Applicable     | 1      | 1        | 1          | 0     | 0      | 0       | 1     | 0   | 0     | 0          | 0     | 0       | 1   |

**Figure 2.2 Snapshot of the Cleaned Dataset**

Note: I did the cleaning in SQL (check file).

For the data cleaning, I started by creating a staging table from the raw data to preserve the original records and corrected the data types for each column at this stage. I then standardized the job titles by categorizing them into 10 simplified titles using a CASE statement, carefully arranging the conditions to avoid misclassification. The salary estimate column was parsed using nested SUBSTRING\_INDEX functions to extract separate minimum and maximum salary columns, then averaged as the midpoint. For company names, I used REGEXP to detect and strip ratings that were appended to the name. The location and headquarters columns were split into separate city and state columns using a comma delimiter, with full state names converted to abbreviations and invalid values like United States set to NULL. I also corrected two headquarters records with an incorrect state code of 061 to NY. Beyond standard cleaning, I extracted 13 binary skill columns from the job description using LIKE pattern matching, stored as integers for easier aggregation, and added two columns of my own, senior to flag senior-level roles and same\_state to flag if the job location matches the headquarters state before dropping the job description column from the final table since the skills had already been extracted from it.

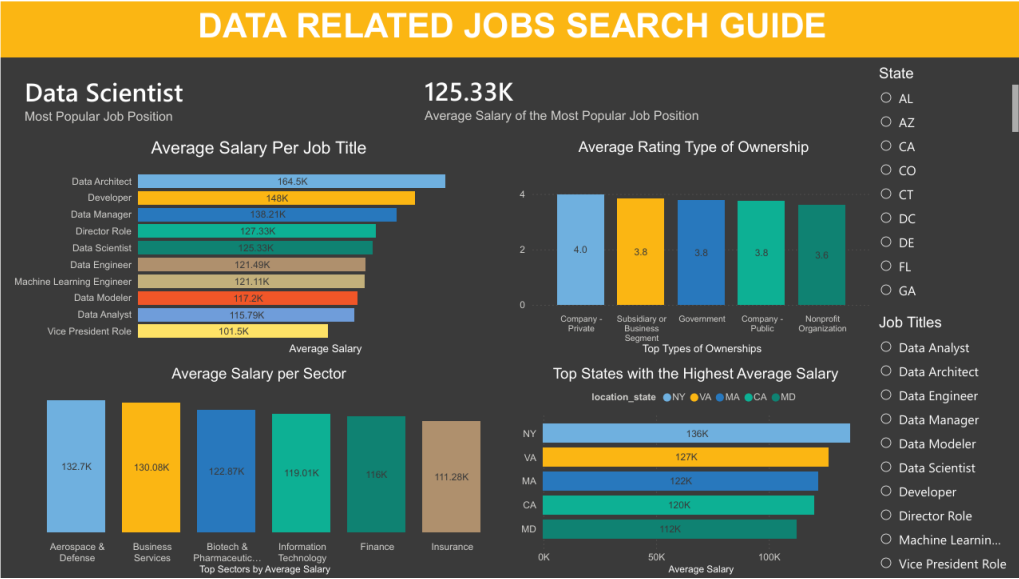


Figure 3.1 Data Science Jobs Dashboard Page 1

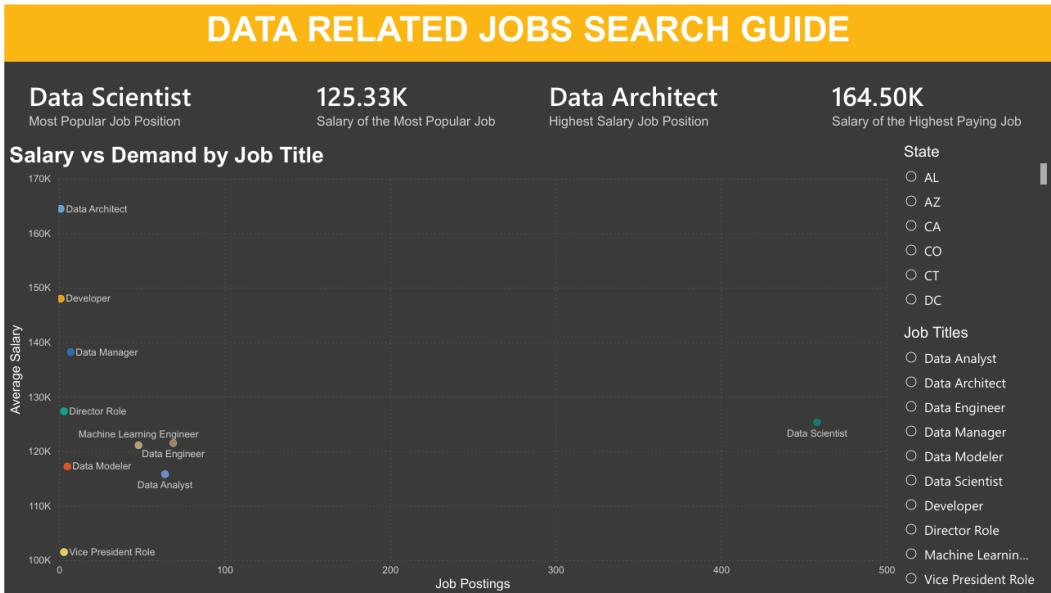
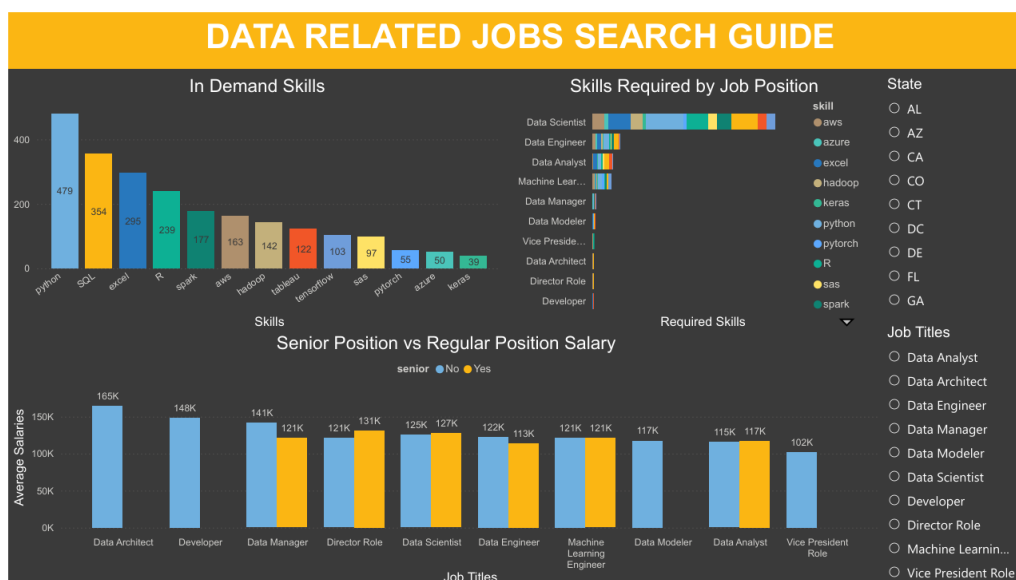


Figure 3.2 Data Science Jobs Dashboard Page 2



**Figure 3.3 Data Science Jobs Dashboard Page 3**

Note: Dashboard screenshots are shown in the Power BI file (check file).

For the EDA, I tried to answer common questions that a job seeker or recruiter in the data field would want to know. The dataset covers data science and data-related job postings from Glassdoor in the United States.

The most posted job title in this dataset is Data Scientist with 458 postings, which is expected given the title of the dataset. As for average salary, Data Architect takes the top spot at \$164,500 average salary, while Vice President Role has the lowest at \$101,500. The salary spread across job titles is relatively wide, which tells us that the specific role matters more than just being in the data field.

For companies, Maxar Technologies has the most job postings with 12 listings. I also did a follow-up analysis comparing companies with high postings against their recorded revenue. I noticed some companies with high postings show Unknown revenue, which could be a red flag for job seekers since it is harder to assess the financial stability of the company.

When comparing highest rated companies (5.0) versus lowest rated companies (2.0), I found that the lowest rated group actually shows a higher average salary. However, this result should be taken with caution since there is only 1 company in the lowest rated group compared to 25 in the highest rated group. Rather than adjusting the range to add more companies to the lowest group, I kept it as is to maintain the integrity of the data and avoid misrepresenting the insight.

The hotspot for job postings by city is San Francisco, CA with 58 postings, followed by other major tech hubs. At the state level, California leads with 155 postings. For average salary by state, New York tops the list at \$136,000, followed by Virginia at \$127,000.

I compared US local companies (headquarters state is a 2-letter state code) against international companies (headquarters state is a longer value indicating a foreign location). US local companies have significantly more postings at 596 versus only 43 for international.

International companies do show a higher average salary, but the small sample size makes this comparison less reliable.

For company characteristics, I looked at size, revenue, type of ownership, and founding year in relation to salaries. Across all four, I found no clear pattern where any one factor strongly influences salary. Company size does not seem to matter, revenue brackets show mostly similar averages with one outlier in the \$50 to \$100 million range which I left as is since there is no way to explain it from this dataset alone. Type of ownership and founding year also show no strong salary influence. The insight here is that for data-related jobs, salary seems to be driven more by the specific role and location rather than company characteristics.

The sector with the highest job demand is Information Technology, which makes sense given the nature of data science roles. However, the sector with the highest average salary is Aerospace and Defense at \$132,700, and with 46 job listings it has enough volume to be considered a reliable figure.

For type of company, I defined better as the combination of highest average rating and highest average salary. Company - Private leads with a 4.0 average rating and \$121,868 average salary, and it also has the most job postings making it the most reliable reference. A hospital type appears at the top for ratings, but with only 1 posting it is not a reliable data point.

Two additional analyses came from the columns I added during cleaning. For senior roles, the data shows that being in a senior position does not clearly result in a higher salary across the board. Some roles like Director Role and Data Scientist even show higher average salaries for non-senior positions, which is a counterintuitive finding worth highlighting. For same state analysis, jobs where the location state matches the headquarters state show a slightly lower average salary compared to companies hiring outside their home state, with a difference of about \$10,206. This could suggest that companies hiring across state lines are offering more competitive packages.

## **Conclusion**

To conclude this Data Analysis project, I can say that the data science job market in this dataset is competitive and varied. Python, SQL, and Excel dominate as the most in-demand skills, and California is clearly the hotspot for job postings. Data Architect has the highest salary while Data Scientist is the most posted role, meaning high demand does not always mean the highest pay. Company characteristics like size, revenue, and founding year do not strongly influence salary, while location and job title seem to matter more. The senior title flag also shows that seniority does not guarantee a salary bump across all roles, which is a useful insight for job seekers negotiating their compensation.

I also want to note that this dataset is a real-world scrape from Glassdoor and not a synthetic one, which means the insights reflect actual market conditions at the time of collection. I made use of tools like SQL for data cleaning and EDA, Power Query in Power BI for data transformation, and Power BI for dashboard creation to complete this project.